

SymPy Bug Card

Bug 54: singularities misses the pole of gamma(x) at zero

Task. Find the complex singularities of the gamma function.

$$\text{singularities}(\Gamma(x), x, \mathbb{C})$$

SymPy's answer.

$$\text{singularities}(\Gamma(x), x, \mathbb{C}) = \emptyset$$

Correct answer. The result must contain 0; in fact, $\Gamma(x)$ has finite poles at $0, -1, -2, \dots$

Explanation. SymPy itself evaluates $\Gamma(0)$ as complex infinity, computes residue 1 at 0, and gives the Laurent expansion $\Gamma(x) = 1/x - \gamma_E + O(x)$. The nonzero $1/x$ term proves a genuine pole, so $\emptyset$ is mathematically false.

Likely root cause. The scanner in `sympy/calculus/singularities.py` handles powers and a small list of logarithmic or inverse-hyperbolic atoms, but it has no path for finite poles of $\Gamma(g(x))$ and does not use gamma's numeric pole knowledge. This is a new failure mode with no public precedent.

Artifact (GitHub).

[bug-report/bug-054-singularities-misses-the-pole-of-gamma-x-at-zero/](#)